

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Tuesday 9 January 2024

Morning (Time: 40 minutes)

Paper reference **31617H/1C**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Chemistry

SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and Functions of Cells and Tissues (Biology)
 - Section B: Periodicity and Properties of Elements (Chemistry)
 - Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- The periodic table of elements can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☐ and then mark your new answer with a cross ☐.

- 1 (a) Figure 1 shows some information about an atom of carbon.

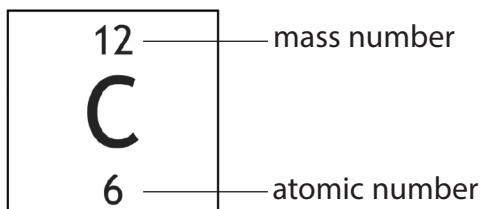


Figure 1

The atom contains six electrons in orbitals around the nucleus.

- (i) Which row of the table shows the correct number of neutrons and protons in the nucleus of the carbon atom?

(1)

		neutrons	protons
A	☐	6	6
B	☐	6	12
C	☐	12	6
D	☐	12	12

- (ii) Complete the electronic configuration for carbon.

(1)

$1s^2 2s^2$

- (b) Carbon dioxide and methane are compounds containing carbon.

- (i) State **one** use of carbon dioxide.

(1)



Methane has the formula CH₄.

(ii) Draw the 3D line diagram for methane.

(1)

Table 1 shows the bond energy of a C—H bond in methane and a C=O bond in carbon dioxide.

bond	bond energy (kJ/mol)
C—H	414
C=O	805

Table 1

(iii) Using the information in Table 1, explain why the C=O bond is stronger than the C—H bond.

(4)

(Total for Question 1 = 8 marks)



P 7 7 2 5 5 A 0 3 1 2

2 (a) The elements in group 7 are known as the halogens.

(i) Which of these halogens is the most reactive?

(1)

- ☐ **A** astatine
- ☐ **B** bromine
- ☐ **C** chlorine
- ☐ **D** iodine

Table 2 shows the boiling points of four halogens.

element	boiling point (°C)
fluorine	–188
chlorine	–35
bromine	59
iodine	184

Table 2

(ii) Explain the trend in the boiling points of the halogens in Table 2.

(3)

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(b) Chlorine reacts with hydrogen to form hydrogen chloride.

(i) Draw the dot and cross diagram for a molecule of hydrogen chloride.

(2)

(ii) Explain why a molecule of hydrogen chloride is polar.

You may draw a diagram to support your answer.

(2)

(Total for Question 2 = 8 marks)



3 Iron is a metal found in the d-block of the periodic table.

(a) Which row of the table is correct for iron?

(1)

		ductility	melting point
A	☐	high	high
B	☐	high	low
C	☐	low	high
D	☐	low	low

Iron can be extracted from iron oxide.

(b) Give the oxidation number of iron in Fe and in Fe_2O_3 .

(2)

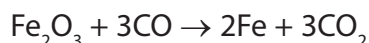
oxidation number of iron in Fe =

oxidation number of iron in Fe_2O_3 =



- (c) A sample of iron oxide was reduced with excess carbon monoxide.

The theoretical yield of iron from this sample is 27.90 g.



- (i) Calculate the minimum mass of iron oxide that was used.

(relative atomic mass: Fe = 55.8, relative formula mass: $\text{Fe}_2\text{O}_3 = 159.6$)

(3)

Show your working.

minimum mass of iron oxide = g

The actual yield of iron produced from this sample was 19.53 g.

- (ii) Calculate the percentage yield of iron.

(2)

Use the equation

$$\text{percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

percentage yield of iron = %

(Total for Question 3 = 8 marks)



4 A teacher reacted a small piece of lithium with water containing universal indicator.

The teacher repeated the reaction using sodium and potassium, using fresh water and universal indicator each time.

A learner observed the reactions and recorded the results in Table 3.

metal	observation
lithium	- fizzing - metal floats and moves around the surface of the water - universal indicator turns purple
sodium	- rapid fizzing - metal floats and moves quickly around the surface of the water - universal indicator turns purple
potassium	- very rapid fizzing - lilac flame and popping sound - metal floats and moves very quickly around the surface of the water - universal indicator turns purple

Table 3

Explain the similarities and differences between these three reactions.

(6)

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(Total for Question 4 = 6 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



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